

Hotel Booking – Distributed Application

Group Members

- Bubacarr Samba
- Anura Shantha H A K
- Nguyen An Ninh

Project Overview

Hotel booking is a blockchain-based application that provides a secure, transparent, and decentralized booking system. The main purpose of this project is to demonstrate how blockchain technology and smart contracts can improve trust, reduce fraud, and simplify the booking process compared to traditional booking systems.

In many traditional booking systems, issues such as manipulation of booking transactions, lack of transparency, delayed confirmation of money transfer, and security vulnerabilities can undermine the credibility of payment. Through the booking D-App, we aim to design a digital booking platform where payments from bookings are securely stored on the blockchain and cannot be altered once submitted. This creates a tamper-proof, transparent, and trustworthy booking process.

The system will include simple features that represent the main activities and users involved in the booking process. The Hotel booking allows the user to

- Check how many rooms are left to book
- If the room is available for booking
- Book a room (reserve the room)
- Check out a booked room (release the room for another booking)

Technical plan

To easily control the transaction on the chain, we decided to rely on a local blockchain network. It gives us massive ETH for transactions and payments. It also rapidly executes, processes, and stores transactions on the chain. To set up a local blockchain, we need to install some components on our computer, these are:

- **Foundry:** A modern, fast, and developer-friendly toolkit for building, testing, and deploying smart contracts on Ethereum and other EVM-compatible blockchains.
- **Anvil:** An Ethereum node designed specifically for development. It is part of the Foundry ecosystem and is built using Axum, a modern web application framework in Rust. Anvil is tailored for creating and testing smart contracts on any EVM-compatible blockchain, providing developers with a streamlined, efficient environment.
- **MetaMask:** A free, open-source, non-custodial hot wallet designed primarily for the Ethereum blockchain and compatible networks. It acts as a gateway to Web3,

enabling users to store, send, receive, and interact with Ether (ETH), ERC-20 tokens, NFTs, and decentralized applications (dApps) directly from their browsers or mobile devices. MetaMask will act as a middle layer between the application and the blockchain network

Once we have set up the environment, we can start building the application, which includes:

- A Solidity contract.
- A web-based application could be built with ReactJS or another popular JavaScript framework.
- Integration between the web app and the Solidity contract.